

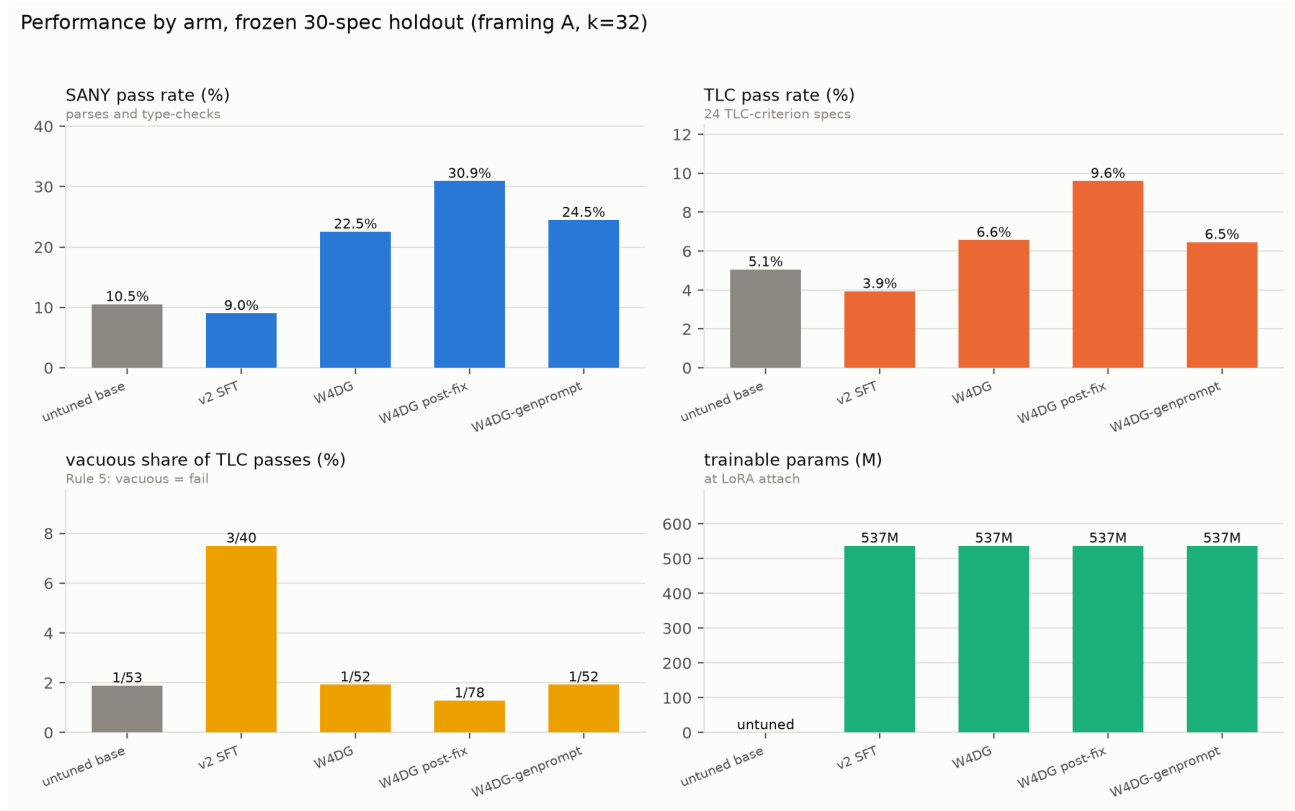
Fine-tuning a TLA+ prover: every run and every measurement, 2026-03 to 2026-08

2026-08-12

Figures regenerate from `tools/brief_figures.py`. Gate-2 rates are recomputed from `results/runs/*/rows.jsonl` at build time: (`spec`, `sample`) deduplicated keep-first, corruption rows dropped, `api_error` excluded from the denominator as an unmeasured draw. RL-era training metrics are read from TLA-Prove logs surviving at git e79a250 and 511a492.

Performance by arm

Performance by arm, frozen 30-spec holdout (framing A, k=32)



arm	SANY	TLC	vacuous	params	per-sample	pass@32
untuned base	10.5%	5.1%	1/53	untuned	6.9%	12/30
120b						
v2 SFT	9.0%	3.9%	3/40	536.8M	5.1%	11/30
W4-diamond-gold	22.5%	6.6%	1/52	536.8M	11.1%	15/30
W4DG post	30.9%	9.6%	1/78	536.8M	13.6%	16/30
prompt-fix						
W4DG-genprompt	24.5%	6.5%	1/52	536.8M	8.1%	12/28

arm	SANY	TLC	vacuous	params	per-sample	pass@32
mech composition	—	—	—	536.8M	pending	pending

vacuous is vacuous draws over TLC-passing draws checked; **params** is trainable parameters at LoRA attach.

All four 120b arms attached **identical** LoRA geometry: 536,813,568 trainable parameters, 0.4574%, 144 expert tensors, **target_modules** q/k/v/o plus **target_parameters** on **mlp.experts.gate_up_proj** and **down_proj**. Verified from each run's own attach-time log line. Trainable capacity is therefore constant across every arm in this table.

verdict_of is population-aware, so the TLC column excludes 6 of 30 specs: 41, 86, 105 and 183 are LIBRARIES (SANY alone is the criterion) and 131, 142 are PROOF_MODULES (TLAPS). The TLC denominator is the remaining 24 specs.

Vacuity is only computed on a TLC-passing draw, so the denominator is small (40–78 draws per arm) and the counts are shown rather than percentages alone.

The RL-era runs are not on this chart. They were evaluated on a different holdout (Diamond-30) under a four-shot repair loop, and their **sany_valid** field is a residual category rather than a pass rate — **holdout_repair.json** records **sany_valid=0** alongside 9 TLC-gold results. Those numbers are in the Diamond-30 section below.

Every run

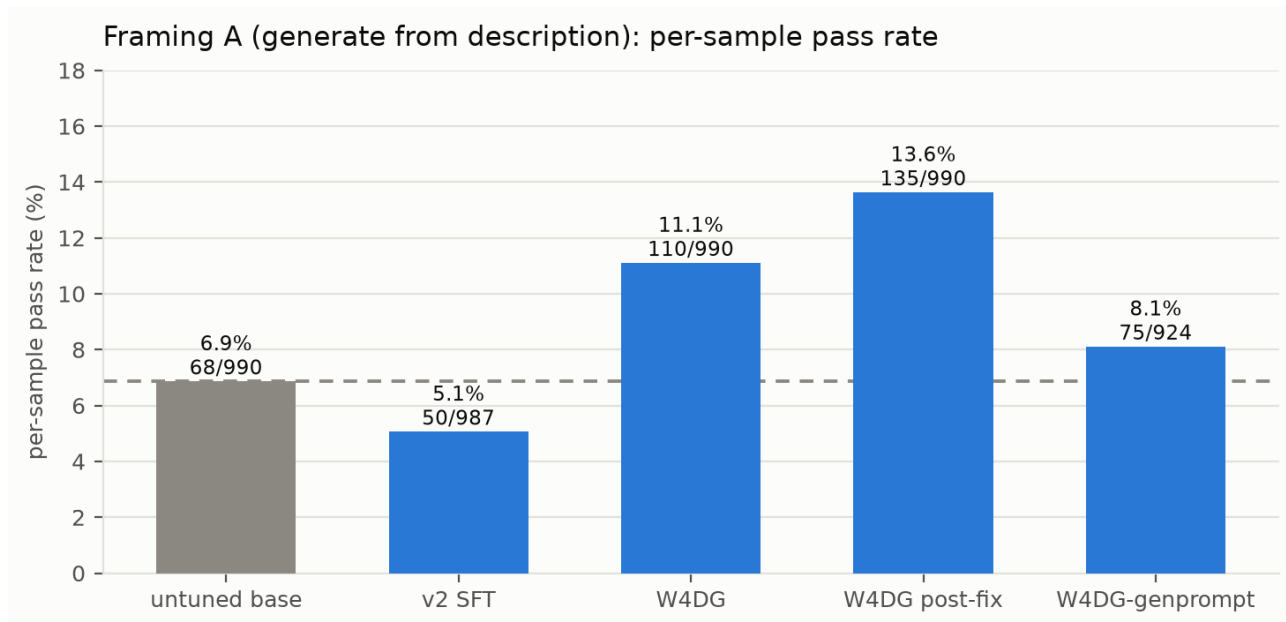
#	date	run	base	corpus	outcome
1	03-22→04-06	rl_loop, 266 cycles	20b	augmented, append-only	SANY 80%→65%, TLC 10%→25%
2	04-03	DPO v13	20b	17 preference pairs	9/20 SANY, 5/20 TLC (v11: 6/20, 2/20)
3	~04-10	piecewise DPO	merged v14	curriculum pairs	loss 0.6661 vs ln2 0.6931; acc 0.60
4	04-11	full-spec GRPO	post-DPO	398 prompts	172 steps; holdout 4/30, 1-shot 1/30
5	04-12	repair GRPO R1	post-DPO	ralph repair pairs	965 steps; holdout 9/30, 1-shot 3/30
6	04-13	repair GRPO R2	post-DPO	R2 harvest	600 steps; holdout 6/30, 1-shot 1/30
7	04-14	repair GRPO R3	—	—	aborted: 152 pairs vs floor 300
8	07-01	repair GRPO retry	20b	35 rows	89 steps zero reward; 1/7 rows; reverted
9	07-12	v2_sft1	20b	39 plain-text pairs	0/10; 82% unextractable
10	07-13	v2_sft2	20b	260 harmony pairs	2/30 pass@4 — first passes recorded
11	07-14	v2_sft2	120b	260	A 5.1%, 11/30; B 26.5%, 18/23
12	07-25	W2.6 repair-v1	20b	508 repair triples	B 9/23 pass@1, 9/23 pass@4
13	07-29	W4-diamond-gold	120b	3,534 rows, bare	A 11.1%, 15/30
14	07-30	W4DG, post prompt-fix	120b	same	A 13.6%, 16/30
15	08-04	W4DG-genprompt	120b	4,219 rows, aligned	A 8.1%, 12/28; B 20.9%, 17/23
16	08-12	mech composition	120b	6,906 (4,119 gen + 2,787 repair)	training; eval pending

Runs 1–8 are LUC-AI4FM/TLA-Prove on 2× RTX 8000 49GB. Runs 9–16 are prove-TLA on Argonne Sophia,

8× A100. A further arm, Qwen3.6-27B dense, is retracted: a gpt-oss-specific `target_parameters` selector matched nothing on a dense model, leaving the FFN at 0.0195% trainable while the job exited 0.

Framing A — generate a spec from a description

30-spec frozen holdout, k=32, temp 0.8.



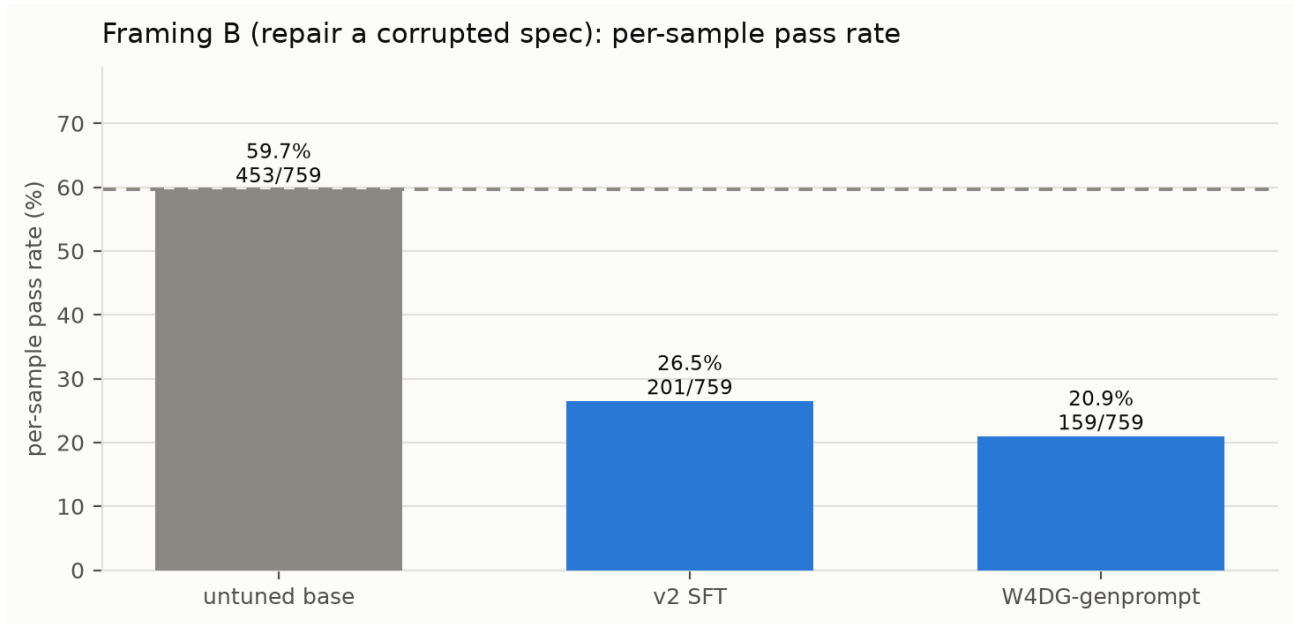
arm	per-sample	pass@32	rows scored	api_error
untuned base 120b	6.9% (68/990)	12/30	990	0
v2 SFT	5.1% (50/987)	11/30	987	3
W4-diamond-gold	11.1% (110/990)	15/30	990	0
W4DG post prompt-fix	13.6% (135/990)	16/30	990	0
W4DG-genprompt	8.1% (75/924)	12/28	924	66

Paired two-level bootstrap, restricted to the 17 specs whose `prompt_sha256` is byte-identical across arms: W4DG vs base **+0.086**, **p=0.063**; W4DG-genprompt vs base **-0.018**, **p=0.66**. The `required_signature` fix changed 13 of 30 framing-A prompts on 2026-07-29, so rows either side of that date are not directly comparable outside the 17-spec matched set.

W4DG-genprompt lost specs 183 and 191 in full (66 `api_error` rows) to a serve walltime expiry; its denominator is 28 specs.

Framing B — repair a corrupted spec

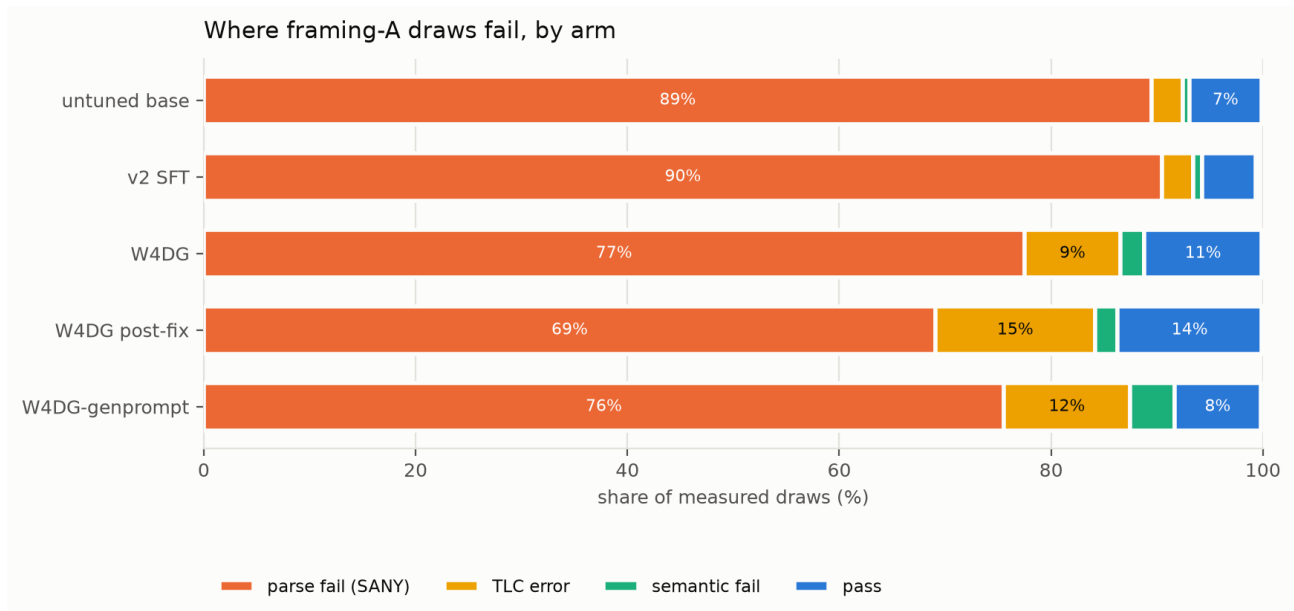
23-spec denominator, k=33.



arm	per-sample	pass@32
untuned base 120b	59.7% (453/759)	21/23
v2 SFT	26.5% (201/759)	18/23
W4DG-genprompt	20.9% (159/759)	17/23

W4DG framing B reached 110 of 990 rows before the serve died and is not scored here.

Where the draws fail



Verdict buckets: parse fail = fail:sany=*; TLC error = fail:tlc=error and fail:tlc=timeout; semantic fail = fail:tlc=fail_invariant, fail_deadlock, fail_liveness.

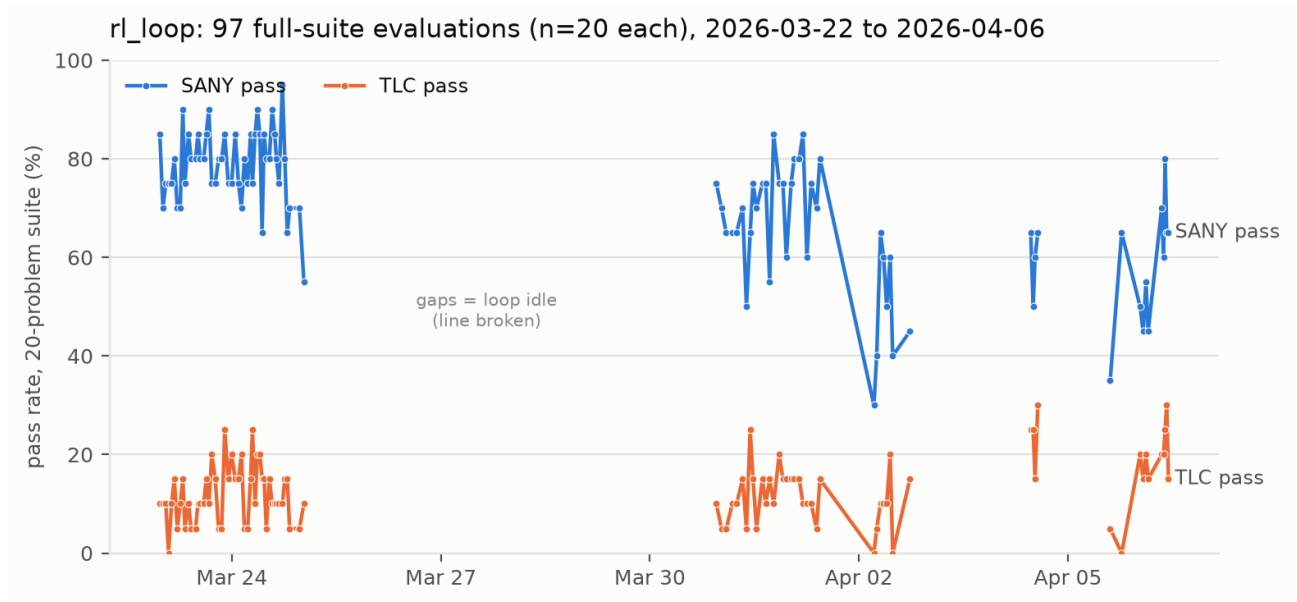
arm	parse fail	TLC error	semantic fail	pass
untuned base	89.5%	2.9%	0.6%	6.9%

arm	parse fail	TLC error	semantic fail	pass
v2 SFT	90.5%	2.9%	0.8%	5.1%
W4-diamond-gold	77.5%	9.1%	2.2%	11.1%
W4DG post prompt-fix	69.1%	15.1%	2.1%	13.6%
W4DG-genprompt	75.5%	11.9%	4.2%	8.1%

Rows do not sum to 100%: 0.1–0.7% per arm is `fail:tlaps=error` or an unmatched verdict. `fail:sany=fail_missing_module` is counted under parse fail.

rl_loop, 266 cycles

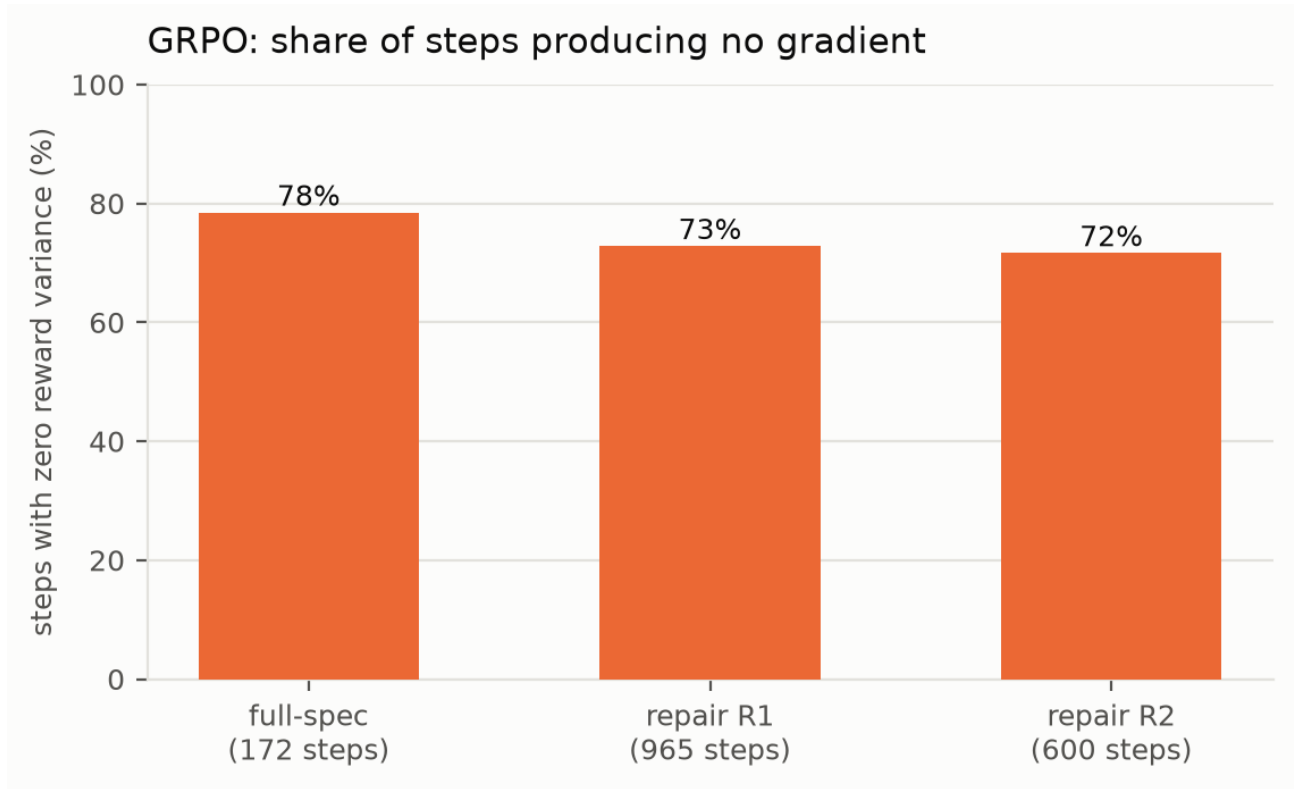
Computed from all 97 full-suite CSVs (n=20 each) in `outputs/benchmark_results`. Quick-eval CSVs (n=12, 107 files) are excluded.



All 228 CSVs record the model as `chatt1a:20b`, a mutable Ollama tag rewritten on each redeploy; no CSV is attributable to a specific checkpoint. The projection in `README_RL_SETUP.txt` was 85% SANY and 40–50% TLC after ~15 cycles.

GRPO reward variance

`frac_reward_zero_std == 1` means every completion in the sampled group scored identically, so the advantage and the update are both zero.



run	steps	mean reward	zero-var steps	KL	entropy
full-spec	172	0.0511	135/172 (78%)	0.002–0.014	0.44 → 0.32
repair R1	965	0.1487	703/965 (73%)	0.002–0.014	0.003–0.3
repair R2	600	0.0930	430/600 (72%)	—	—

Repair R1’s mean reward of 0.1487 equals the shaping function’s hard-coded “no change” constant of 0.15 (`repair_reward.py:70`). At temperature 0.5 the July run emitted byte-identical completions within a group. No reward hacking and no verifier false positives were found in any run.

Diamond-30 holdout, RL era

Three numbers rest on this holdout. The 9/30 is published as 30% Gold (arXiv:2606.06133; AUDIT.md:12; docs/formallm.md:25).

eval	fixed	single-shot	timeouts	tiers
full-spec GRPO	4/30	1/30	2	4 gold, 5 silver
repair R1	9/30	3/30	4	9 gold
repair R2	6/30	1/30	1	6 gold, 2 silver

Three properties of this holdout, each verified directly:

- All 30 holdout module names appear in `data/diamond_gen_topics.json`. `fullspec_dataset.py` defaults `include_topics=True`, `train_rl_fullspec.py` passes it, and the loader implements no holdout filter. The SFT path does filter by module name (`train.py:110`).
- The eval is a four-shot repair loop with verifier feedback at temperatures 0.50, 0.70, 0.90. Single-shot counts above are rows with `result == fixed` and `attempts_used == 1`.
- `repair_pipeline.log` records `LoRA merge FAILED` at 00:31:19; the next line starts the eval against Ol-lama tag `chattla:20b-repair`. No log shows that tag rebuilt from the R1 adapter.

Corpora

corpus	rows	liveness	top family	comment density	used by
v2 harmony	260	—	—	—	runs 10, 11
W2.6 repair triples	508	—	—	—	run 12
W4-diamond-gold	3,534	6.2%	mutex_locks 35.8%	—	runs 13, 14
W4DG 5010-render	4,119	12.4%	mutex_locks 35.5%	3.9% of lines; 38.7% of targets carry 1+ comment	run 15 base
W4DG-genprompt	4,219	—	—	—	run 15
mech composition	6,906	—	—	—	run 16

W4 corpus mutation-catch rate: **12.7%** real (69% of survivors have no mutation site the battery can kill).
Public TLA+ liveness rate for reference: 23–34%.

Oracle repair pairs in run 16: 4,000 survivors attempted, 2,787 kept (1,981 SANY-class, 806 TLC-class, 69.7% yield). Each pair is a corrupted survivor whose corruption was executed against SANY/TLC and kept only if it actually fails, with the original Opus-teacher text as the target.

Defects found

#	layer	defect	exit code	status
1	training	<code>target_parameters</code> hardcoded for gpt-oss; dense FFN frozen at 0.0195%	0	fixed + aborting floor
2	training	<code>is_moe()</code> read only top-level config; Qwen3.5-35B-A3B's 256 experts resolved as dense	0	fixed
3	training	PBS echoed <code>TRAIN_EXIT=\$?</code> then exited 0	0	fixed, 18 scripts
4	training	<code>Mxfp4Config</code> passed by family not checkpoint; transformers 5.12.1 refuses	1	fixed
5	training	2-GPU <code>CUDA_VISIBLE_DEVICES</code> hardcode	—	fixed
6	training	staging tree held stale <code>lora_resolver.py</code> / <code>train.py</code>	—	fixed, verified per-run
7	eval	<code>required_signature</code> dropped cfg substitution RHS; 13/30 framing-A prompts under-specified	—	fixed 07-29
8	eval	serve <code>--max-model-len</code> 4096 truncated 4.4k–17k-token repair prompts	—	fixed, run quarantined
9	eval	<code>summary.json</code> written from in-memory rows, understated resumed runs	—	ledger-only scoring
10	eval	<code>api_error</code> rows counted as done on resume	—	open, bitten twice
11	eval	pass@32 collapses 32 draws to 1 bit	—	per-sample now primary

Four of the six training defects exited 0 on runs that had not trained what they were supposed to train.

Not yet tried

lever	status	cost
mechanics supervision	run 16, training now	spent
grammar-constrained decoding	2 EBNF grammars scaffolded (<code>tla_module_v0.ebnf</code> , <code>tla_proof_v0.ebnf</code>); no decoder wired	no training compute
multi-task split (generation / repair / abduction / fuzzing)	verifiers exist for all four; never trained jointly	1 train run
RL restart	trainers reusable; blocked on <code>frac_reward_zero_std</code>	1 measurement, then 1 run
holdout widening past 30 specs	effects below ~4 points unresolvable at n=30	amendment + decontam pass

Open measurements

- Run 16 eval: framings A and B, k=32.
- W4DG-genprompt specs 183 and 191: 66 rows unmeasured.
- W4DG framing B: 880 of 990 rows unmeasured.
- Difficulty probe: 436 of 2,400 rows short; measures pass rate under the training prompt rather than the eval prompt.